DSA REFERENCE

Trie Problem Ladder

A sequenced practice ladder from warm-up to interview stretch.

Store strings by shared prefixes so lookup cost depends on key length rather than number of stored keys.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

DSA / TRIE

DSA

Trie Problem Ladder

CORE IDEA

Store strings by shared prefixes so lookup cost depends on key length rather than number of stored keys.

Recognition cues

+ prefix queries
+ dictionary word search
+ autocomplete
+ many string lookups with shared prefixes

Implementation sequence

+ choose child representation
+ mark terminal nodes
+ separate exact and prefix search
+ consider memory tradeoffs

Invariant

The path from root to a node spells one prefix; an end marker distinguishes full words from prefixes.

Worked reasoning

Autocomplete follows the query prefix, then enumerates terminal descendants subject to a result limit.

Complexity

O(L) insert and lookup for key length L; memory is proportional to stored prefix nodes.

Failure modes

- [ ] forgetting end markers
- [ ] treating prefix as a full word
- [ ] oversized fixed alphabets
- [ ] unbounded enumeration after prefix match

Practice ladder

Trie Problem Ladder: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Implement Trie**	State the cue, invariant, edge case, and proof before implementation.
2	**Design Add and Search Words**	State the cue, invariant, edge case, and proof before implementation.
3	**Word Search II**	State the cue, invariant, edge case, and proof before implementation.
4	**Replace Words**	State the cue, invariant, edge case, and proof before implementation.
5	**Search Suggestions System**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.